

Lecture Plan

Name of Course	Master of Computer Applications	
AKTU Course Code	09	
Branch	MCA	
Semester	2nd	
Section	A	
Total Number of Students	26	
Name of Faculty	Dr. Om Prakash Kushwaha	
Number of Lecture Proposed	48	
Subject Name	Web Technology	Subject Code: BMC 201

S.No	Unit No	Topic	CO	No of Lectures Required	Actual Date of Completion	Suggested Reference
1	1	Introduction and Web Development Strategies, History of Web and Internet, Protocols Governing Web	1	1		
2		HTML-Introduction, HTML Tags, HTML-Grouping Using Div & Span		1		
3		HTML-Lists, HTML-Images, HTML-Hyperlink, HTML Table		1		
4		HTML-iframe, HTML-Form		1		
5		Introduction of CSS, CSS Syntax, External Style Sheet using < link > ,		1		
6		Multiple Style Sheets, Value Lengths and Percentages, CSS-Selectors,		2		
7		CSS-Box Model, Floats,		2		
8		Clear, Introduction to Bootstrap.		1		
No of Lectures Required to complete Unit 1				10		
9	2	Introduction to JavaScript, Creating Variables in JavaScript,	2	1		
10		Creating Functions in JavaScript,		1		
11		UI Events, Returning Data from Functions,		2		
12		Working with Conditions, looping in JavaScript,		1		
13		Block Scope Variables		1		
14		Working with Objects,		1		
15		Creating Object using Object Literals,		1		
16		Manipulating DOM Elements with JavaScript		1		
No of Lectures Required to complete Unit 2				09		

17		Servlet Overview and Architecture, Interface Servlet		1		
18		Servlet Life Cycle		1		
19		Handling HTTP get Requests,		1		
20		Redirecting Requests to Other Resources, Session Tracking,		1		
21		Cookies, Session Tracking with Http Session.		1		
22		Java Server Pages (JSP): Introduction, Java Server Pages Overview, A First Java Server Page Example		2		
23		Implicit Objects, Scripting, Standard Actions,		2		
24				Directives, Custom Tag Libraries.		1
No of Lectures Required to complete Unit 3				10		
25	4	Spring Core Basics-Spring Dependency Injection concepts, ,		1		
26		Introduction to Design patterns, Factory Design Pattern,		1		
27		Strategy Design pattern, Spring Inversion of Control,		2		
28		AOP, Bean Scopes Singleton		1		
29		Prototype, Request, Session,		1		
30		Application, WebSocket,		2		
31		Auto wiring, Annotations,		1		
32		Life Cycle Call backs, Bean Configuration styles		2		
No of Lectures Required to complete Unit 4				10		
33	5	Spring Boot- Spring Boot Configuration	5	1		
34		Spring Boot Annotations, Spring Boot Actuator,		1		
35		Spring Boot Build Systems, Spring Boot Code Structure,		2		
36		Spring Boot Runners, Logger,		1		
37		BUILDING RESTFUL WEB SERVICES,		1		
38		Rest Controller, Request Mapping,		1		
39		Request Body, Path Variable, Request Parameter,		1		
40			GET, POST, PUT, DELETE APIs, Build Web Applications		1	

No of Lectures Required to complete Unit 5	09
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Text Books & References	
1	Burdman J., "Collaborative Web Development – Strategies and Best practices for Web Teams", Addison-Wesley.
2	Xavier C, "Web Technology & Design", New Age international Publishers.
3	Bayross I., "Web Technologies", BPB Publications.
4	Schildth H., "The Complete Reference – HTML & CSS", McGraw Hill.
5	Bakliwal S., "Hands-on Application Development using Spring Boot", BPB Publications.

Course Outcomes (COs)		
At the end of this course students will demonstrate the ability to:		
CO 1	Apply the knowledge of HTML and CSS to develop web application and analyze the insights of internet programming to implement complete application over the web.	K3, K6
CO 2	Understand, analyze and apply the role of JavaScript in the workings of the web and web applications.	K2, K3
CO 3	Understand, analyze and build dynamic web applications using servlet and JSP.	K2, K3
CO 4	Develop Spring-based Java applications using Java configuration, XML configuration, annotation-based configuration, beans and their scopes, and properties.	K2, K4, K6
CO 5	Develop web application using Spring Boot and RESTful Web Services.	K3, K6

Name of Faculty with Signature: Dr. Om Prakash

Lecture Plan In-charge: Mr. Praveen Kr. Tripathi

Head of Department: Mr. Prashant Shrivastava

Dean Academics: Dr. Kamal Prakash Pandey